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MILITARY SPECIFICATION

PROJECTILE, 4.2 INCH, ILLUMINATING, M335A2 METAL PARTS ASSEMBLY

This specification is approved for use for all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Projectile, 4.2 Inch, Illuminating, M335A2 Metal Parts Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC 1315

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Frankford Arsenal, ATTN: SARFA-MDM, Phila, PA 19137 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1 Cont'd

DRAWINGS

Munitions Command

F10534155	- Projectile, 4.2 Inch, Illuminating, M335A2 Metal Parts Assembly
F8848553	- Pallet
LI-10534155	- Index of Inspection Equipment Lists for Projectile, 4.2 Inch, Illuminating, M335A2 Metal Parts Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI-B46.1	- Surface Texture

(Applications for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018.)

ASTM E-8	- Tension Testing of Metallic Materials
ASTM E-10	- Brinell Hardness of Metallic Materials
ASTM B-117	- Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Phila, PA 19103.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications, and the following:

3.2 Heat treatment or stress relief.-Heat treatment or stress relief applied to any component part shall be uniform throughout and shall be so conducted as to produce in each component the mechanical properties specified on the applicable drawing.

3.2.1 Hot forge heat treat method.

3.2.1.1 Body.-Prior to finish machining, projectile bodies manufactured from forgings shall be subjected to heat treatment to attain the required mechanical properties. The heat treatment shall consist of heating above the

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3.2.1.1 (Cont'd)

critical temperature, not to exceed 1600 degrees Fahrenheit (^{o}F), followed by an oil quench. After quenching, the body forging shall be tempered without delay at a temperature above 750 degrees Fahrenheit, suitable to produce required mechanical properties. After removal from the tempering furnace, the body forgings shall be slowly cooled in air (forced air permitted) to a temperature of not more than 700 degrees Fahrenheit, after which cooling may be accelerated by water spray.

3.2.1.2 Base plug and nose adapter.-The base plug and nose adapter may be used in the forged condition provided that mechanical property requirements are met. If base plug and nose adapter forging do not meet the required uniform mechanical properties as forged, heat treatment by quenching and tempering or by normalizing is required. When normalizing the components shall be heated to a temperature of 1600 degrees Fahrenheit, minimum, held at temperature for a minimum of 30 minutes and cooled in such a manner as to produce uniform mechanical properties throughout each component.

3.2.2 Cold shaping and hot cup - cold draw methods.-Cold working shall be so accomplished as to produce in the cold worked components the mechanical properties specified. The components shall be stress relieved at a temperature of 700 degrees Fahrenheit, minimum, for a minimum of 30 minutes following the final forming operation during which time plastic flow takes place.

3.2.3 Tubing.-Bodies manufactured from tubing shall be stress relieved in accordance with 3.2.2.

3.3 Mechanical properties.-Mechanical properties as specified on the applicable component drawing shall apply throughout the component.

3.4 Metal defects.

3.4.1 Components (body, adapter and base plug).-The components, including the cavity, shall be free from cracks, splits, bursts, cold shuts, pipe, porosity, inclusions, folds, seams and other metal defects.

3.4.2 Components (body, adapter, and base plug).-The component and their cavities shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, pits and sharp edges.

3.4.3 Visual standards.-Visual standards for metal defects of paragraph 3.4.2 will be established for each contract after the first month's production, or whenever requested by the procuring activity (see 6.2).

3.5 Joint quality.-The brazed joint shall be free of entrapped slag, voids, unwetted or unbonded areas and other brazing defects. The aggregate area of these defects shall not exceed 25 percent of the area of the faying surfaces; and the maximum extent of an unbonded surface including solder ring groove, shall not exceed 50 percent of the overlap distance (width) of the joint along any longitudinal line.

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3.6 Proving ground functional test.-The projectile assembly shall reveal no evidence of failure, when tested at excess pressure. Metal parts failure shall be:

a. Loss or break-up of metal parts components, except base plug.

3.7 Workmanship.-The requirements for workmanship are as shown on the applicable drawing, in the applicable specification and the following:

3.7.1 Painting.-Painting shall comply with the requirements of the applicable drawing. All paint shall be dry to the touch before packing for shipment. Drying time prior to testing shall be in accordance with the applicable specification.

3.7.2 Surface finish.-The requirements for surface finish are as detailed on the applicable drawing.

3.7.3 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as specified on the applicable drawing.

3.7.4 Burr.-No part shall have a burr which might interfere with the assembly of the round or which might be injurious to personnel handling the item.

3.7.5 Foreign matter.-No part or assembly shall contain dirt, grease, chips, rust, corrosion or other foreign matter. Particular attention should be given to the cavity of the body assembly, to insure that no grit, scale, accumulated braze alloy residual flux or other foreign matter remains therin after final cleaning.

3.7.6 Protective coating.-The protective coating shall be in accordance with the applicable drawing and the specification listed herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

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4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with specification MIL-A-2550, except as specified herein.

4.1.3 Inspection criteria.

a. Acceptance of all inspections required by the drawings and specification shall be considered as a condition precedent to performing the Proving Ground Functional test (see 3.6).

b. Acceptance or rejection of a lot by the Government shall be based on results of the Proving Ground functional tests.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of thirty-five projectile metal parts as follows:

25 Projectile metal parts complete (painted)
5 Projectile metal parts complete (unpainted)
5 Sets of component parts unassembled and unpainted

Twenty of the above completed and painted metal parts assemblies shall be forwarded to a designated proving ground for functional testing. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials including packaging and packing shall be the same as used in regular production and shall be obtained from the same source of supply. A first article sample or portion thereof as directed by the procuring agency shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing processes, materials used, drawing, specification or source of supply, such as to significantly affect product uniformity as determined by the responsible technical agency.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

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4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Heat treatment lots.-Heat treatment lots shall contain component forgings from not more than one heat of steel that have been heat treated in the same heat treatment equipment, at the same temperature, in one unchanged process. The heat treatment lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed (see 6.3).

4.3.1.3 Stress relieved lots.-Stress relieved lots shall contain cold worked components from not more than one heat of steel that have been cold worked by the same process, utilizing the same tooling and equipment and stress relieved in the same furnace at the same temperature, at the same length of time. The stress relieved lot thus constituted shall be the basis for selection of samples for the tension test hereinafter prescribed (see 6.3).

4.3.1.4 Tubing lot.-Components manufactured from tubing shall be fabricated from tubing containing not more than one heat of steel. Tubing lots thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed (see 6.3).

4.3.1.5 Bar stock lot.-Components machined from bar stock shall contain material from not more than one heat of steel and produced in one unchanged process. Whenever heat treatment is applicable (whether during manufacture of bar stock or manufacture of component, the heat treatment shall be performed in the same heat treating equipment at the same temperature in one unchanged process. The machining lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed (see 6.3).

4.3.1.6 Miscellaneous lots.-Small lots (such as components from abnormally small heats of steel or components that have become separated from previously delivered lot, etc.) may at the discretion of the supplier be grouped into miscellaneous heat treatment, stress relieved or machining lots as the case may be. Such lots shall be identified by a miscellaneous lot code number and processed in accordance with either 4.3.1.2, 4.3.1.3, or 4.3.1.4, as applicable. The miscellaneous lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.7 Projectile lots.-When it has been determined that bodies of a heat treated or stress relieved lot have complied with the mechanical properties requirement of the applicable drawing, they may be grouped for further processing.

4.3.1.8 Shipping lots.-Shipping lots shall consist of projectile assemblies produced by one supplier, under one contract, in one unchanged process, in accordance with the same drawings, revisions, under conditions of a continuous production, free of interruptions other than those due to the end of shift day, or work week. The shipping lot thus constituted shall be the basis for the selection of samples for the proving ground functional test. Lot size shall be determined by the procuring contracting officer.

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4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as cracks, burrs and foreign matter. The roughness comparison specimens prescribed by ASTM B46.1 shall be used as a basis for surface roughness determination.

4.3.2.1 Tool control dimensions.-

a. Dimensions marked tool control shall be gaged at the beginning or end of each working day (24 hrs), during First Article Inspection and for the first component after each tooling change, adjustment or repair. Where destruction of the components is necessary to gage such dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production.

b. When the contractor has satisfied (a) above, visual inspection of tool control dimensions will be permitted. Visual inspection shall be performed every two (2) hours for each machine in operation. If any tool control dimension(s) is in doubt during visual inspection, the dimension(s) shall be immediately gaged (see "a" above) to determine acceptability. Corrective action will immediately be instituted by the contractor whenever a defective tool control dimension is found.

4.3.2.2 Spacer, Drawing C10534156 covering a detail of Drawing F10534155.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Diameter of force fit surface	Gage
102.	Length of force fit surface, min., if applicable	Gage
103.	Surface finish on force fit surface	Gage
104.	Diameter of holes	Gage
MINOR		
201.	Diameter outside, large, if applicable	Gage
202.	Length, total	Gage
203.	Location of holes	Gage
204.	Thickness of wall	Gage
205.	Radii or chamfers missing or incorrect	Visual (See 6.1)
206.	Surface finish, improper (except as otherwise classified)	Visual (See 6.1)
207.	Burr	Visual
208.	Foreign matter	Visual

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4.3.2.2.1 Spacer alternative, Drawing D10551970, covering a detail of Drawing F10534155.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Diameter of force fit surface	Gage
102.	Ovality of force fit surface	Gage
103.	Length of force fit surface, max.	Gage
104.	Surface finish on force fit surface	Gage
105.	Diameter of holes	Gage
106.	Diameter of fuze clearance cavity	Gage
107.	Length from fuze clearance cavity to rear	Gage
MINOR		
201.	Diameter outside, large	Gage
202.	Diameter outside, small	Gage
203.	Length from front shoulder to rear	Gage
204.	Length from front face to large outside diameter, min.	Gage
205.	Length from front face to small outside diameter, max.	Gage
206.	Location of holes	Gage
207.	Thickness of wall	Gage
208.	Width and depth of force fit surface undercut, max.	Gage
209.	Radii or chamfers, missing or incorrect	Visual (See 6.1)
210.	Suface finish improper (except as otherwise classified)	Visual (See 6.1)
211.	Burr	Visual
212.	Foreign matter	Visual

4.3.2.3 Adapter, Nose, Drawing D10534158, covering a detail of Drawing F10534155.

CRITICAL		
	None defined	
MAJOR		
101.	Runout of spacer cavity with brazing diameter and seating shoulder	Gage
102.	True position of small inside diameter with spacer cavity	Gage

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4.3.2.3 (Cont'd)

	Defect	Method of Inspection
103.	Diameter of brazing surface	Gage
104.	Diameter of spacer cavity	Gage
105.	Depth of force fit diameter	Gage
106.	Surface finish force fit diameter	Gage

MINOR

201.	Diameter of seating shoulder	Gage
202.	Diameter of outside small, rear	Gage
203.	Length of brazing surface	Gage
204.	Length, total	Gage
205.	Presence, size and location of groove (if applicable)	Visual (See 6.1)
206.	Presence, size and location of undercut rear of seating shoulder (if applicable)	Visual (See 6.1)
207.	Radii or chamfers missing or incorrect	Visual (See 6.1)
208.	Surface finish improper (except as otherwise classified)	Visual (See 6.1)
209.	Burr	Visual
210.	Foreign matter	Visual

4.3.2.4 Adapter assembly, Drawing D10534157, covering a detail of Drawing F10534155.

CRITICAL

None defined

MAJOR

101.	Lenth, total, max.	Gage
102.	Gap between spacer and adapter shoulder	Visual

MINOR

None defined

4.3.2.5 Plug, Base, Drawing D10534160, covering a detail of Drawing F10534155 (before paint).

CRITICAL

None defined

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4.3.2.5 (Cont'd)

	Defect	Method of Inspection
MAJOR		
101.	Runout of large outside diameter with force fit surface and front face of flange	Gage
102.	Runout of pitch diameter of thread with force fit surface and front and rear face(s) of flange	Gage
103.	True position of rotating disc seat with major diameter of thread	Gage
104.	Diameter of force fit surface	Gage
105.	Diameter of flange	Gage
106.	Height of rotating disc seat (rear shoulder)	Gage
107.	Thickness through flange and rotating disc seat	Gage
108.	Width of rotating disc seat	Gage
109.	Finish on force fit surface improper	Visual (See 6.1)
MINOR		
201.	Diameter of thread undercut	Gage
202.	Width of thread undercut	Gage
203.	Thickness between force fit surface and cavity	Gage
204.	Depth of cavity	Gage
205.	Height of thread boss	Gage
206.	Length of taper	Gage
207.	Length, total	Gage
208.	Radii or chamfers missing or incorrect	Visual (See 6.1)
209.	Surface finish, improper	Visual (See 6.1)
210.	Presence, size and location of undercut on force fit surface (if applicable)	Visual (See 6.1)
211.	Burr	Visual (See 6.1)
212.	Foreign matter	Visual

4.3.2.6 Plug base, Drawing D10534160, covering a detail of Drawing F10534155 (after paint).

CRITICAL		
	None defined	
MAJOR		
101.	Pitch diameter of thread, min.	Gage
102.	Pitch diameter of thread, max.	Gage
103.	Major diameter of thread, min.	Gage
104.	Force fit surface damaged	Visual (See 6.1)

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4.3.2.6 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Paint on force fit surface	Visual
202. Protective coating inadequate or damaged	Visual
203. Wax emulsion missing from indicated surfaces	Visual
204. Corrosive preventive compound missing from thread	Visual
205. Foreign matter	Visual

4.3.2.7 Body, Drawing D10534159, covering a detail of Drawing F10534155.

CRITICAL	
None defined	
MAJOR	
101. Diameter of brazing surface	Gage
102. Length to basic diameter	Gage
103. Length to rear of forward bourrelet	Gage
104. Length to front of rear bourrelet	Gage
105. Thickness of wall at ogive	Gage
106. Variation of wall thickness at ogive	Gage
MINOR	
201. Diameter at front, outside	Gage
202. Length, total	Gage
203. Surface finish, improper	Visual (See 6.1)
204. Burr	Visual
205. Foreign matter	Visual

4.3.2.7.1 Body alternative, Drawing D10551972, covering a detail of Drawing F10534155.

CRITICAL	
None defined	
MAJOR	
101. Runout of force fit diameter with bourrelet	Gage
102. Diameter of force fit surface	Gage

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4.3.2.7.1 (Cont'd)

	Defect	Method of Inspection
103.	Length to outside basic diameter	Gage
104.	Length to inside basic diameter	Gage
105.	Length from front face to bottom of force fit diameter	Gage
106.	Thickness of wall in ogive, min.	Gage
107.	Variation in wall thickness	Gage

MINOR

201.	Length from front face to rear of thread undercut	Gage
202.	Diameter of thread undercut	Gage
203.	Length, total	Gage
204.	Radii or chamfers missing or incorrect	Visual (See 6.1)
205.	Surface finish improper	Visual (See 6.1)
206.	Burr	Visual
207.	Foreign matter	Visual

4.3.2.8 Metal Parts Assembly, Drawing F10534155 (before paint).

CRITICAL

None defined

MAJOR

101.	Runout of pitch diameter of nose thread with bourrelets and front face of nose	Gage
102.	Runout of nose adapter with bourrelets	Gage
103.	Runout of forward ogive with bourrelets (if applicable)	Gage
104.	Runout of rear ogive with bourrelets	Gage
105.	Runout of body with bourrelets	Gage
106.	Runout of illuminant charge cavity with bourrelets	Gage
107.	Runout of force fit cavity at rear with bourrelets	Gage
108.	Perpendicularity of rear face with bourrelets	Gage
109.	Diameter of bourrelets, min.	Gage
110.	Diameter of body	Gage
111.	Diameter of illuminant charge cavity	Gage
112.	Depth of thread cavity	Gage
113.	Length from front face to rear of forward bourrelets (if applicable)	Gage
114.	Length from front face to front of rear bourrelet (if applicable)	Gage
115.	Overlap between body and nose adapter, joint improper	Visual (See 6.1)
116.	Surface finish of force fit cavity at rear	Visual (See 6.1)

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4.3.2.8 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Diameter of nose outside	Gage
202. Diameter of thread undercut (if applicable)	Gage
203. Width of thread undercut (if applicable)	Gage
204. Surface finish improper	Visual (See 6.1)
205. Radii or chamfers missing or incorrect	Visual (See 6.1)
206. Burr	Visual
207. Foreign matter	Visual

4.3.2.9 Metal Parts Assembly, Drawing F10534155 (after paint).

CRITICAL	
1. Pressure stamp missing (indicating failure to pass pressure test) (if applicable)	Visual
MAJOR	
101. Pitch diameter of nose thread, max.	Gage
102. Pitch diameter of nose thread, min.	Gage
103. Minor diameter of nose thread, max.	Gage
104. Diameter of bourrelets, max.	Gage
105. Diameter of force fit cavity at rear	Gage
106. Depth of force fit cavity at rear	Gage
107. Length of effective nose thread, min. (if applicable)	Gage
108. Length from front face to rear of spacer, max. (if applicable)	Gage
109. Damage to force fit cavity at rear	Visual (See 6.1)
MINOR	
201. Protective coating inadequate or damaged	Visual
202. Thread(s) damaged or not full	Visual
203. Wax emulsion inadequate or damaged from indicated areas	Visual
204. Corrosion preventive compound missing from nose thread	Visual
205. Marking missing, incorrect or illegible	Visual
206. Foreign matter	Visual

4.3.2.10 Pallet Drawing F8848553.

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4.3.2.10 (Cont'd)

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Pallet damaged	Visual
202. Sleeve missing	Visual
203. Improperly packed	Visual
204. Strapping missing, loose or broken	Visual and Manual

4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Metal defects.-Each metal parts assembly and base plug shall be visually examined to determine compliance with 3.4.1 and 3.4.2. Failure to comply with the requirement of 3.4.1 and 3.4.2 shall be cause for rejection.

4.3.3.2 Mechanical properties.

4.3.3.2.1 Hardness. (Body only)

4.3.3.2.1.1 Heat treatment lots.-In order to determine uniformity of heat treatment, each body in the heat treatment lot (as defined in 4.3.1.2) shall be subjected to a Brinell hardness test or any method of hardness testing approved by the procuring activity.

4.3.3.2.1.2 Stress relieved lots.-Twenty-five cold shaped bodies from the first batch of each stress relieved lots (as defined by 4.3.1.3) processed through the stress relief furnace and 5 bodies from each one hours production thereafter for the balance of the stress relieved lot shall be subjected to a Briness hardness test, or any method of hardness testing approved by the procuring activity to assess uniformity of the furnace.

4.3.3.2.1.3 Miscellaneous lots.-Miscellaneous heat treatment lots shall be subjected to a hardness test in accordance with 4.3.3.2.1.1. Miscellaneous stress relieved lots shall be subjected to a hardness test in accordance with 4.3.3.2.1.2.

4.3.3.2.2 Tension test.

4.3.3.2.2.1 Body.

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4.3.3.2.2.1.1 Heat treatment lots.-Two projectile bodies, one of the hardest and one of the softest as determined by the hardness test shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. At the option of the supplier, a production control sample may be utilized to establish a hardness range. When a production control sample is used, the hardness range thus established shall be maintained throughout the heat treatment lot. The size of the sample shall be determined by the contractor. Projectile bodies which fall outside of the established range may be re-heat treated to fall within the range or new mechanical properties may be determined from newly selected projectile bodies which will represent the true extremes of hardness in the heat treatment lot.

4.3.3.2.2.1.2 Stress relieved lots.-Two projectile bodies, one of the hardest and one of the softest, as determined by the hardness test from the first twenty-five projectiles shall be subjected to this test. Failure of any tension test specimen to comply with the mechanical properties requirement of the applicable drawing shall be cause for rejection of the lot. When it has been determined that the mechanical properties comply and furnace temperature and time have been established, the furnace temperature and time shall be maintained throughout each stress relief lot. The furnace temperature may vary within twenty-five degrees Fahrenheit but not below the minimum temperature of 700 degrees Fahrenheit and the minimum 30 minutes soak time. Furnace temperaure and time will be recorded every hour to assess adequate control of stress relief equipment. The furnace temperature will be calibrated each eight hours by an independent method.

4.3.3.2.2.2 Base plugs and nose adapters.-Two base plugs and two nose adapters from a heat treatment, stress relieved lot as defined by 4.3.1.2 or 4.3.1.3 shall be subjected to this test to determine compliance with the yield strength and elongation requirements of the applicable drawing. Failure of any tension test specimen to comply shall be cause for rejection of the lot.

4.3.3.2.2.2.1 Bar stock lots.-When base plugs or nose adapters are machined from bar stock in such a manner that the mechanical properties are nor altered, test coupons may be removed from one end of each 3 hours for each heat of steel in the lot and tested in lieu of the component. Failure of any tension test specimens to comply with the yield strength and elongation requirements of the applicable drawing shall be cause for rejection of the bar stock lot.

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4.3.3.2.2.3 Tubing lot.-When the nose adapters and spacers are machined from tubing in such a manner that the mechanical properties are not alteres, test coupons may be removed from one end of each of 3 tubes for each heat of steel and tested in lieu of testing the component. Failure of any tension test specimens to comply with the yield strength and elongation requirements of the applicable component drawing shall be cause for rejection of the tubing lot.

4.3.3.2.2.4 Miscellaneous lots.-Samples for tension test of miscellaneous heat treatment, stress relieved, tubing, bar stock or forging lots shall be selected in accordance with 4.3.3.2.2.1 thru 4.3.3.2.2.3 as applicable. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. Tension tests shall be performed on each miscellaneous heat of steel before being grouped into a miscellaneous lot.

4.3.3.3 Brazed joint.

4.3.3.3.1 Strength of brazed joint.-Each body assembly shall be subjected to a mechanical load test to determine compliance with requirements of the applicable drawing. Failure of the assembly to comply shall be cause for rejection of the assembly. Body assemblies which comply with drawing requirements shall be stamped with the letter "P" as indicated on the assembly drawing.

4.3.3.3.2 Joint quality.-After pressure testing but prior to the application of protective coating, 2 body assemblies from each lot shall be selected at random, sectioned and separated to determine compliance with paragraph 3.5. Failure of any specimen to comply shall be cause for rejection of the lot.

4.3.3.4 Cleaning and application of protective coating.-The sample sizes and acceptance criteria for the inspection of cleaning the item and the application of protective coating shall be as specified in the applicable specification recommended herein.

4.3.3.4.1 Salt spray.-The sample size and acceptance criteria for salt spray testing shall be in accordance with the applicable specification. Failure to comply with the requirements of the applicable drawing shall be cause for taking the action recommended by the applicable specification.

4.3.4 Proving ground functional tests.-The following proving ground tests will be performed by the Government.

4.3.4.1 First article sample.

4.3.4.1.1 Initial production sample.-The quantities listed in Table I shall be ballistically tested in the phases indicated.

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4.3.4.1.1 (Cont'd)

TABLE I

Phase	Quantity
Excess pressure	20

4.3.4.2 Excess pressure.-Samples provided in accordance with Table I will be subjected to this test. Failure of any projectile assembly to comply with 3.6 shall be cause for rejection of the initial production sample.

4.3.5 Inspection equipment.-Index of inspection equipment lists LI-10534155 identifies the inspection equipment required to perform the examinations and tests prescribed herein.

4.4 Test methods and procedures.

4.4.1 Mechanical properties.

4.4.1.1 Hardness.

4.4.1.1.1 Body.-After final heat treatment or stress relief, but prior to finish machining the hardness test shall be performed on the rear bourrelet approximately 3 inches from the rear face or at any other location on the body necessary for process control. The equipment and procedures for performing this test are prescribed by ASTM Method E-10.

4.4.1.2 Tension test.

4.4.1.2.1 Body.-From the samples provided by the heat treat or stress relief lots, two tension test specimens conforming to the largest obtainable standard "R" type specimen as prescribed by ASTM Method E-8 shall be taken longitudinally from opposite sides of the body 180 degrees apart. The number of specimens and theor location are for the convenience of the contractor and Government and shall not restrict the contractor or Government from testing additional specimens at other locations to determine that the mechanical properties of the projectile body meet the drawing requirements. The criteria for performing this test shall be as prescribed by ASTM Method E-8; except that elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0 percent).

4.4.1.2.2 Base plug.-From the samples supplied in accordance with 4.3.3.2.2.2, one tension test specimen conforming to the largest obtainable standard "R" type specimen as prescribed by ASTM Method E-8 shall be taken in a transverse direction .25 to .35 inches from the centerline of the base plug. The criteria for performing this test shall be measured to the nearest thousandth (0.000) prior to conversion into percent

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4.4.1.2.2 (Cont'd)

elongation. Percent elongation shall be recorded to the nearest tenth (0.0 percent).

4.4.1.2.3 Nose adapter.-From the samples supplied in accordance with 4.3.3.2.2.2, one tension test specimen conforming to the largest obtainable standard "R" type specimen as prescribed by ASTM Method E-8 shall be taken longitudinally, from the adapter. The criteria for performing this test shall be as prescribed by ASTM Method E-8; except that elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0 percent).

4.4.1.2.4 Machined from the bar stock.-From the test coupons supplied in accordance with 4.3.3.2.2.2.1, one tension test specimen conforming to an "R" type specimen as prescribed by ASTM Method E-8 shall be taken longitudinally from the center of each test coupon. The criteria for performing this test shall be as prescribed by ASTM Method E-8; except that elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0 percent).

4.4.1.2.5 Nose adapter and spacers machined from tubing.-From the test coupons supplied in accordance with 4.3.3.2.2.3, one tension test specimen conforming to the largest obtainable standard "R" type specimen for the nose adapter and one tension test specimen conforming to the largest obtainable standard "T" specimen for the spacer shall be taken longitudinally from each test coupon as prescribed by ASTM Method E-8. The criteria for performing this test shall be as prescribed by ASTM Method E-8; except that elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0 percent).

4.4.1.3 Strength of brazed joint.-The projectile shall be placed over the holding fixture and the pressure applied to the ogive at the rear of the brazed joint. Care shall be exercised in applying the pressure to assure that the fixture contacts the ogive below the brazed joint.

4.4.1.3.1 Joint quality.-The samples selected in accordance with Para. 4.3.3.3.2 shall be sectioned and separated as follows: (a) the forward portion of the projectile assembly, 1-inch minimum rear of the nose adapter, shall be removed in such a manner that no portion of the brazed joint will be damaged, (b) the forward portion of the body containing the nose adapter shall then be cut vertically through the center of the nose adapter, (c) each portion of the body shall be peeled from the nose adapter section. After peeling, the brazed area of the two halves of the nose adapter and the mating pieces of the body shall be visually examined to determine compliance with 3.5.

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4.4.2 Protective coating.-The procedures for testing the cleaning and surface treatment methods for components and assemblies shall be in accordance with the methods and criteria for testing as specified in the applicable specification.

4.4.2.1 Salt spray.-The salt spray test prescribed by the applicable drawing shall be performed utilizing the equipment and procedures specified by ASTM Method B-117.

4.4.3 Proving ground functional tests.

4.3.3.1 Excess pressure.-The projectiles supplied in accordance with 4.3.4, inert loaded and dummy fuzed, will be fired for recovery, in an acceptable standard gun based upon current gun tube condemnation criteria. A propellant charge adjusted to produce 109 to 117 percent of the upper pressure limit of the propellant used with the applicable projectile. Photographic equipment will be set up near the gun muzzle to record the flight of the projectile. An attempt will be made to recover at least 60 percent of the projectiles fired in this phase for examination to determine compliance with 3.6.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.

5.1.1 Level C.-Metal parts assemblies shall be palletized in accordance with Drawing F8848553 and MIL-STD-1169.

5.1.2 Level C.-Base plugs shall be packed in accordance with MIL-STD-1169.

6. NOTES

6.1 Visual examination qualification.-When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Visual standards.-Visual standards to be established at contractor's plant in accordance with 3.4.5.

6.3 Optional procedure for heat lot identity.-For a period of at least four months of production, the contractor shall maintain heat lot identity as per requirements of item specification. From the data generated during the four month period, the contractor shall establish a reliable hardness range of high and low values which meet the yield strength and elongation requirements of the applicable drawing. During this period and when the contractor changes to a hardness range identity, it behooves the contractor to monitor the chemistry of the heats of steel to assure compliane to the established hardness range. Once the hardness range has been established by the contractor, the following criteria for the heat treatment shall apply:

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6.3 (Cont'd)

a. Heat treatment lots (definition).-Each eight (8) hours of production of body forgings processed through heat treatment furnaces shall be designated as a heat treatment lot. The heat treatment lot thus constituted shall be the basis for selection of samples for tension test hereinafter prescribed.

b. Heat treatment lot hardness test.-Each body in the heat treatment lot as defined in (a) above shall be subjected to a Brinell Hardness Test to verify that each body falls within the established hardness range. Projectile bodies which fall outside the established range shall be rejected.

c. Heat treatment lot tension test.-Two bodies after hardness testing shall be randomly selected by Government personnel from each 8 hours production or fraction thereof and shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for immediate halt of production, and the following screening procedure shall be implemented. All projectile in the production line processed forward of the hardness test shall be screened 100% for hardness and the results recorded. The two highest and the two lowest hardness readings of the screened projectiles shall be subjected to mechanical properties testing. If results are favorable, the contractor may proceed with production in accordance with his established range. If results indicate that the high or low samples do not meet the mechanical properties, the contractor shall establish a new hardness range to assure compliance with the mechanical properties as specified on the applicable drawing. This new hardness range shall be approved by the Government technical agency. The screened projectiles shall be considered suspect and removed to a bond area where mechanical properties testing will be conducted by the contractor with a reduced hardness range as per the recorded values. When the mechanical properties meet the requirements of the drawing at a reduced hardness range, all suspect projectiles falling within this range can be released for further processing. All other projectiles that fall out of this range will be considered rejects.

d. Hardness tester.-Each machine in operation shall be calibrated prior to start of each day's production.

6.4 Steel blooms, billets, slabs, bars and tubing.

6.4.1 It is the responsibility of the manufacturer of the projectile metal parts to select a fully-killed carbon steel of a grade compatible with the abilities of the equipment and the methods of processing the metal within the operations of the shops of the forger and machiner to produce the projectile to the dimensions, mechanical properties and metal integrity specified as an end item requirement.

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6.4.2 The supplier of the selected steel raw material is responsible only for furnishing a metal composition, grade, and type with dimensions, weight, soundness and freedom from metallurgical or mechanical defect limitations identifiable with the standard AISI grade and complying with the specific ASTM designation and tests selected by the projectile manufacturer.

6.4.3 In the application of carbon steel blooms, billets, slabs and bars, forging quality processing techniques adapter to use of the selected steel should be applied with the expectation that there will be variables in the metal cross-sectional dimensions, in weight per unit length, in the hardenability variance from heat to heat, in the mechanical condition of the surface, and in the straightness of the raw material. Economic advantages will be realized if the process and equipment within the shops of the forger and the machiner have been engineered in terms of equipment sequence of operations, tooling designs, handling devices, and processing techniques having the capabilities of compensating for such variables.

6.5 Ordering data.-Procurement documents should specify the title, number and the date of this specification.

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